

SNEHA PANDIT

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EDUCATION

- **GLA University** | *B.Tech in Computer Science & Engineering (Spec. AI & ML)* | *Expected: 2027*
- **Core Coursework:** Advanced Machine Learning, Natural Language Processing, Deep Learning, Data Structures & Algorithms, OS.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, Java, C++.
- **AI/ML & Research:** Multimodal ML, Transformer Models, NLP, Sequence Modeling, Deep Learning, PyTorch, Hugging Face, RoBERTa.
- **Systems & Database:** Node.js, Express, FastAPI, AsyncIO, BullMQ, Redis, Prisma ORM, RESTful APIs, WebSockets, MongoDB, SQLite.
- **Infrastructure & Security:** AES-256 Field-Level Encryption, JWT Authentication, Git, GitHub Actions, Docker, Linux Environments.

RESEARCH & PROJECT EXPERIENCE

Sentia: Multimodal Emotion Drift Analytics Platform | [GitHub](#) | [Report](#) | *Python, PyTorch, RoBERTa, FastAPI, WebSockets*

- Architected a 7-stage parallel inference pipeline (`get_sentia_intelligence`) executing concurrent text and vocal feature extraction loops via a Python `ThreadPoolExecutor` to maintain a strict **P95 latency < 4s**.
- Normalized the 27-class `GoEmotions` dataset (58k Reddit samples) to a canonical 7-class schema, fine-tuning a RoBERTa-base transformer to achieve a macro F1-score above **0.70**.
- Improved multi-turn classification accuracy over a text-only baseline by executing a confidence-weighted late-fusion architecture combining text embeddings with a facial expression CNN trained on FER-2013.
- Tracked temporal emotional instability across multi-turn user sessions using **Total Variation Distance (TVD)** and Euclidean distance shifts across a 7D probability vector space.
- Implemented a rule-based clause segmentation engine to isolate sub-message strings at syntactic conjunction boundaries to detect intra-message emotional transitions.

Panopticon: Distributed Political-Economic Data Observability Engine | [GitHub](#) | [Report](#) | *TypeScript, Node.js, Express, Prisma, BullMQ*

- Engineered a type-safe, high-throughput distributed tracing backend using **TypeScript** and **Prisma ORM** to aggregate and map multi-node political, asset, and economic data models.
- Developed asynchronous, background data-ingestion workers via **BullMQ** to orchestrate pipelines processing **10k+ structured events/day** across task queues and automated cron cycles.
- Integrated an isolated LLM processing layer via background workers to parse unstructured ingested news feeds, perform automated entity extraction, and dynamically update relational data fields.
- Architected a security authorization layer using **JWT authentication** and cryptographic verification to secure backend endpoint entry routes against unauthorized administrative data injection.
- Designed event-driven diagnostic telemetry loops using **asynchronous WebSockets** to broadcast low-latency system-health visualization components and data status flags to a monitoring cockpit.

OPEN-SOURCE CONTRIBUTIONS & LEADERSHIP: Project Admin / Contributor | *GirlScript Summer of Code (GSSoC) | 2025 – 2026*

- **Project Admin (2026):** Moderating repository code health, structuring detailed contribution guidelines, and reviewing/merging contributor Pull Requests across deep learning and backend optimization repositories.
- **Contributor (2025-2026):** Implemented modular feature extensions, resolved Git branch conflicts, and refined processing scripts across open-source AI/ML codebases.
- **Repository Engineering:** Maintained comprehensive public markdown specifications, active roadmap files, and automated GitHub Actions CI/CD workflows to guarantee main branch compilation stability.

AWARDS & ACHIEVEMENTS

- **National Finalist** | *Smart India Hackathon (SIH), Ministry of Education, Govt. of India* | 2024
 - **Project Theme:** Smart Land Governance AI Engine. Selected out of thousands of national teams to compete at the 36-hour Grand Finale; engineered the core asynchronous backend framework and model validation integration layer.
- **University Selection Lead** | *Smart India Hackathon Internal Nominee* | 2024
 - Secured **1st Place** in campus-wide preliminaries out of all competing teams at GLA University, earning the official institutional nomination for the national rounds.
- **National Phase 2 Selection** | *IBM Praambh Hack Challenge* | 2025
 - Advanced through competitive national screening phases evaluating technical system architecture, cloud-instance optimization, and machine learning pipeline scalability.